

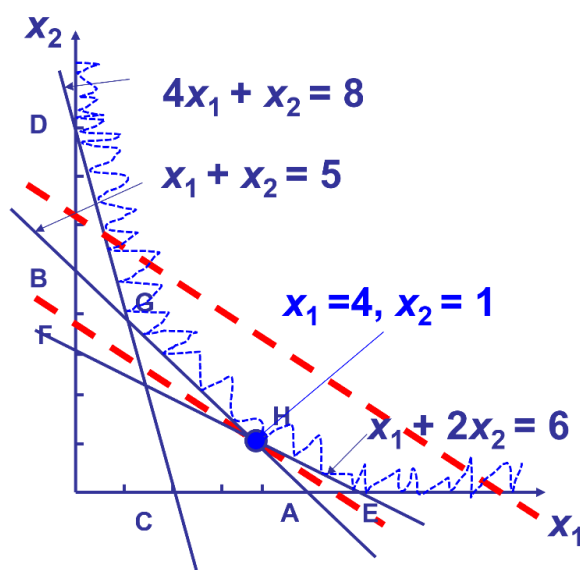
《系统工程与优化方法》习题与解答

线性规划

1. 利用图解法求解下列线性规划问题：

$$\begin{aligned} \min Z &= 2x_1 + 3x_2 \\ \text{s. t. } &\begin{cases} x_1 + x_2 \geq 5 \\ 4x_1 + x_2 \geq 8 \\ x_1 + 2x_2 \geq 6 \\ x_1 \geq 0, x_2 \geq 0 \end{cases} \end{aligned}$$

解：



由上图可知最优解为 $x=[4 \ 1]^T$ ，目标函数最小值为 11。

2. 利用单纯形法求解下列线性规划问题：

$$\begin{aligned} \max Z &= 6x_1 + 5x_2 \\ \text{s. t. } &\begin{cases} x_1 + 3x_2 \leq 90 \\ 2x_1 + x_2 \leq 75 \\ 2x_1 + 2x_2 \leq 80 \\ x_1 \geq 0, x_2 \geq 0 \end{cases} \end{aligned}$$

解：

上述线性规划问题标准型为：

$$\begin{aligned} \max Z &= 6x_1 + 5x_2 \\ \text{s. t. } &\begin{cases} x_1 + 3x_2 + x_3 = 90 \\ 2x_1 + x_2 + x_4 = 75 \\ 2x_1 + 2x_2 + x_5 = 80 \\ x_1, x_2, x_3, x_4, x_5 \geq 0, x_5 \geq 0 \end{cases} \end{aligned}$$

列表如下：

c_j	6			5	0	0	0	θ_i
c_b	x_b	b	x_1	x_2	x_3	x_4	x_5	
0	x_3	90	1	3	1	0	0	$\theta_1=90/1=90$
0	x_4	75	2	1	0	1	0	$\theta_2=75/2=37.5$
0	x_5	80	2	2	0	0	1	$\theta_3=80/2=40$
检验数 σ_j			$\sigma_1=c_1-(c_3a_{11}+c_4a_{21}+c_5a_{31})=6$	$\sigma_2=c_2-(c_3a_{12}+c_4a_{22}+c_5a_{32})=5$	0	0	0	

c_j	6			5	0	0	0	θ_i
c_b	x_b	b	x_1	x_2	x_3	x_4	x_5	
0	x_3	52.5	0	2.5	1	-0.5	0	$\theta_1=52.5/2.5=21$
0	x_1	37.5	1	0.5	0	0.5	0	$\theta_2=37.5/0.5=75$
0	x_5	5	0	1	0	-1	1	$\theta_3=5/1=5$
检验数 σ_j			0	$\sigma_2=c_2-(c_3a_{12}+c_1a_{22}+c_5a_{32})=2$	0	$\sigma_4=c_4-(c_3a_{14}+c_1a_{24}+c_5a_{34})=-3$	0	

c_j	6			5	0	0	0	θ_i
c_b	x_b	b	x_1	x_2	x_3	x_4	x_5	
0	x_3	40	0	0	1	2	-2.5	$\theta_1=52.5/2.5=21$
0	x_1	35	1	0	0	1	-0.5	$\theta_2=37.5/0.5=75$
0	x_2	5	0	1	0	-1	1	$\theta_3=5/1=5$
检验数 σ_j			0	0	0	$\sigma_4=c_4-(c_3a_{14}+c_1a_{24}+c_2a_{34})=-1$	$\sigma_5=c_5-(c_3a_{15}+c_1a_{25}+c_2a_{35})=-2$	

所有检验数均小于 0，因此最优解为 $x^*=[35 \ 5 \ 40]^T$ ，目标函数最大值为 $Z^*=235$ 。

3. 写出下列线性规划问题的对偶问题：

$$\begin{aligned} \min Z &= 6x_1 + 5x_2 \\ \text{s.t.} \quad &\begin{cases} x_1 + 3x_2 \leq 5 \\ 2x_1 + x_2 \leq 75 \\ 2x_1 + 2x_2 \leq 80 \\ x_1 \geq 0, x_2 \geq 0 \end{cases} \end{aligned}$$

解：

对偶问题为：

$$\begin{aligned} \max Z' &= 5y_1 + 75y_2 + 80y_3 \\ \text{s.t.} \quad &\begin{cases} y_1 + 2y_2 + 2y_3 \geq 6 \\ 3y_1 + y_2 + 2y_3 \geq 5 \\ y_1 \geq 0, y_2 \geq 0 \end{cases} \end{aligned}$$

4. 已知下列线性规划问题的对偶问题的最优解为 $y^*=[0, -2]^T$ ：

$$\begin{aligned} \min Z &= 2x_1 - x_2 + 2x_3 \\ \text{s.t.} \quad &\begin{cases} -x_1 + x_2 + x_3 = 4 \\ -x_1 + x_2 - x_3 \leq 6 \\ x_1 \leq 0, x_2 \geq 0 \end{cases} \end{aligned}$$

其中, x_3 无约束。求原问题最优解。

解:

原问题的标准型为:

$$\begin{aligned} \min Z &= 2x_1 - x_2 + 2x_3 \\ \text{s. t. } &\begin{cases} -x_1 + x_2 + x_3 + x_4 = 4 \\ -x_1 + x_2 - x_3 + x_5 = 6 \\ x_1 \leq 0; x_2, x_5 \geq 0; x_4 = 0 \end{cases} \end{aligned}$$

原问题的对偶问题为:

$$\begin{aligned} \max Z' &= 4y_1 + 6y_2 \\ \text{s. t. } &\begin{cases} -y_1 - y_2 \geq 2 \\ y_1 + y_2 \leq -1 \\ y_1 - y_2 = 2 \\ y_2 \leq 0 \end{cases} \end{aligned}$$

其中, y_1 无约束。化为标准型:

$$\begin{aligned} \max Z' &= 4y_1 + 6y_2 \\ \text{s. t. } &\begin{cases} -y_1 - y_2 - y_3 = 2 \\ y_1 + y_2 + y_4 = -1 \\ y_1 - y_2 + y_5 = 2 \\ y_2 \leq 0; y_3, y_4 \geq 0; y_5 = 0 \end{cases} \end{aligned}$$

将最优解 $\mathbf{y}^*=[0, -2]^T$ 代入上述标准型得:

$$y_3 = 0, y_4 = 1$$

由互补松弛定理得:

$$\mathbf{y}^* \mathbf{x}_s = 0 \rightarrow [y_1^* \quad y_2^*] \begin{bmatrix} x_4 \\ x_5 \end{bmatrix} = 0 \xrightarrow{y_1^*=0} x_5 = 0$$

$$\mathbf{x}^* \mathbf{y}_s = 0 \rightarrow [x_1^* \quad x_2^* \quad x_3^*] \begin{bmatrix} y_3 \\ y_4 \\ y_5 \end{bmatrix} = 0 \xrightarrow{\substack{y_3=0 \\ y_4=1 \\ y_5=0}} x_2^* = 0$$

将上述变量值代入原问题约束条件有:

$$\begin{cases} -x_1 + x_3 = 4 \\ -x_1 - x_3 = 6 \end{cases}$$

解上述线性方程组可得 $x_1^* = -5, x_2^* = 0, x_3^* = -1$ 。

无约束优化方法

5. 用最速下降法求函数 $f(x_1, x_2) = 10x_1^2 + x_2^2$ 极小点, 初始点 $\mathbf{x}^{(0)}=[1, 1]^T$ 。

解:

梯度: $\nabla f(x_1, x_2) = [20x_1 \quad 2x_2]^T$

(1) $\nabla f(x_1^{(0)}, x_2^{(0)}) = [20x_1^{(0)} \quad 2x_2^{(0)}]^T = [20 \quad 2]^T$

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} - t_0 \nabla f(x_1^{(0)}, x_2^{(0)}) = [1 \quad 1]^T - t_0 [20 \quad 2]^T$$

$$x_1^{(1)} = 1 - 20t_0, x_2^{(1)} = 1 - 2t_0$$

$$f(x_1^{(1)}, x_2^{(1)}) = 10(1 - 20t_0)^2 + (1 - 2t_0)^2 = 11 - 404t_0 + 4004t_0^2$$

$$\frac{\partial f(x_1^{(1)}, x_2^{(1)})}{\partial t_0} = 8008t_0 - 404 = 0 \rightarrow t_0 = \frac{404}{8008} \approx 0.05$$

$$x_1^{(1)} = 1 - 20t_0 = 0, x_2^{(1)} = 1 - 2t_0 = 0.9$$

(2) $\nabla f(x_1^{(1)}, x_2^{(1)}) = [20x_1^{(1)} \quad 2x_2^{(1)}]^T = [0 \quad 1.8]^T$

$$\mathbf{x}^{(2)} = \mathbf{x}^{(1)} - t_1 \nabla f(x_1^{(1)}, x_2^{(1)}) = [0 \quad 0.9]^T - t_1 [0 \quad 1.8]^T$$

$$x_1^{(2)} = 0, x_2^{(2)} = 0.9 - 1.8t_1$$

$$f(x_1^{(2)}, x_2^{(2)}) = (0.9 - 1.8t_1)^2$$

$$\frac{\partial f(x_1^{(2)}, x_2^{(2)})}{\partial t_1} = 0 \rightarrow t_1 = 0.5$$

$$x_1^{(2)} = 0, x_2^{(2)} = 0 \text{ 为极小点}$$

(3) $\nabla f(x_1^{(2)}, x_2^{(2)}) = [20x_1^{(2)} \quad 2x_2^{(2)}]^T = [0 \quad 0]^T$

$$x_1^{(2)} = 0, x_2^{(2)} = 0 \text{ 为极小点。}$$

6. 用 Newton 法求函数 $f(x_1, x_2) = x_1^2 + 25x_2^2$ 极小点, 初始点 $\mathbf{x}^{(0)} = [2, 2]^T$ 。

解:

梯度: $\nabla f(x_1, x_2) = [2x_1 \quad 50x_2]^T$;

海森矩阵:

$$\nabla^2 f(x_1, x_2) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 50 \end{bmatrix}$$

海森矩阵的逆矩阵:

$$[\nabla^2 f(x_1, x_2)]^{-1} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix} = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.02 \end{bmatrix}$$

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} - [\nabla^2 f(x_1^{(0)}, x_2^{(0)})]^{-1} \nabla f(x_1^{(0)}, x_2^{(0)}) = \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 0.5 & 0 \\ 0 & 0.02 \end{bmatrix} \begin{bmatrix} 4 \\ 100 \end{bmatrix} = 0$$

$$\nabla f(x_1^{(1)}, x_2^{(1)}) = 0$$

因为 $f(\mathbf{x})$ 为二次函数，一步迭代即得平稳点，而且海森矩阵为正定矩阵，所以 $\mathbf{x}^{(1)}$ 为极小点。

7. 用 FR 共轭梯度法求函数 $f(x_1, x_2) = x_1^2 + 2x_2^2$ 极小点，初始点 $\mathbf{x}^{(0)} = [5, 5]^T$ 。

解：

$$\text{梯度: } \nabla f(x_1, x_2) = [2x_1 \quad 4x_2]^T;$$

$$(1) \quad \nabla f(x_1^{(0)}, x_2^{(0)}) = [10 \quad 20]^T, \mathbf{p}^{(0)} = [-10 \quad -20]^T, \mathbf{x}^{(1)} = \mathbf{x}^{(0)} + t_0 \mathbf{p}^{(0)} = \begin{bmatrix} 5 - 10t_0 \\ 5 - 20t_0 \end{bmatrix}$$

$$f(x_1^{(1)}, x_2^{(1)}) = (5 - 10t_0)^2 + 2(5 - 20t_0)^2 = 75 - 500t_0 + 900t_0^2$$

$$\frac{\partial f(x_1^{(1)}, x_2^{(1)})}{\partial t_0} = 1800t_0 - 500 = 0 \rightarrow t_0 = \frac{5}{18}$$

$$\mathbf{x}^{(1)} = \begin{bmatrix} \frac{20}{9} & -\frac{5}{9} \end{bmatrix}^T$$

$$\nabla f(x_1^{(1)}, x_2^{(1)}) = \begin{bmatrix} \frac{40}{9} & -\frac{20}{9} \end{bmatrix}^T$$

$$(2) \quad \alpha_1 = \frac{\|\nabla f(x_1^{(1)}, x_2^{(1)})\|_2}{\|\nabla f(x_1^{(0)}, x_2^{(0)})\|_2} = \frac{4}{81}$$

$$\mathbf{p}^{(1)} = -\nabla f(x_1^{(1)}, x_2^{(1)}) + \alpha_1 \mathbf{p}^{(0)} = \frac{1}{9} \begin{bmatrix} -40 \\ 20 \end{bmatrix} - \frac{4}{81} \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{bmatrix} \frac{-400}{81} \\ \frac{100}{81} \end{bmatrix}$$

$$\mathbf{x}^{(2)} = \mathbf{x}^{(1)} + t_1 \mathbf{p}^{(1)} = \begin{bmatrix} \frac{20}{9} \\ -\frac{5}{9} \end{bmatrix} + t_1 \begin{bmatrix} \frac{-400}{81} \\ \frac{100}{81} \end{bmatrix} = \frac{1}{81} \begin{bmatrix} 180 - 400t_1 \\ 100t_1 - 45 \end{bmatrix}$$

$$\frac{\partial f(x_1^{(2)}, x_2^{(2)})}{\partial t_1} = \left[\frac{1}{81} (180 - 400t_1) \right]^2 + 2 \left[\frac{1}{81} (100t_1 - 45) \right]^2 = 0 \rightarrow t_0 = \frac{5}{18}$$

$$\frac{\partial f(x_1^{(2)}, x_2^{(2)})}{\partial t_1} = 3020t_1 - 1359 = 0 \rightarrow t_1 = \frac{1359}{3020} = \frac{9}{20}$$

$$\mathbf{x}^{(2)} = \mathbf{x}^{(1)} + t_1 \mathbf{p}^{(1)} = \begin{bmatrix} \frac{20}{9} \\ -\frac{5}{9} \end{bmatrix} + \frac{9}{20} \begin{bmatrix} \frac{-400}{81} \\ \frac{100}{81} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\nabla f(x_1^{(2)}, x_2^{(2)}) = 0$$

因此, $\mathbf{x}^{(2)} = [0 \ 0]^T$ 为极小点。